Course : Data Science

Lecture On: NLP Course 4 Revision

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#LifeKoKaroLift

Data Science Certification Program

Today’s Agenda

1 Lexical Processing

2 Syntactic Processing

3 Semantic Processing

NLP → Lexical
NLP → Syntactic
NLP → Semantic

Lexical Processing (NLP)

→ lexicon → word

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* Processing at the words (tokens) level.
* Sequence of words does not matter. (Drawbacks)

S1 ≈ S2

$\left.\begin{array}{l}\text{S1 :- Rama Killed Ravana} \\ \text{S2 :- Ravana Killed Ram}\end{array}\right\}$ lexical processing

* Here, we do not worry about the context of the sentence.

Text Preprocessing

→ tokenisation

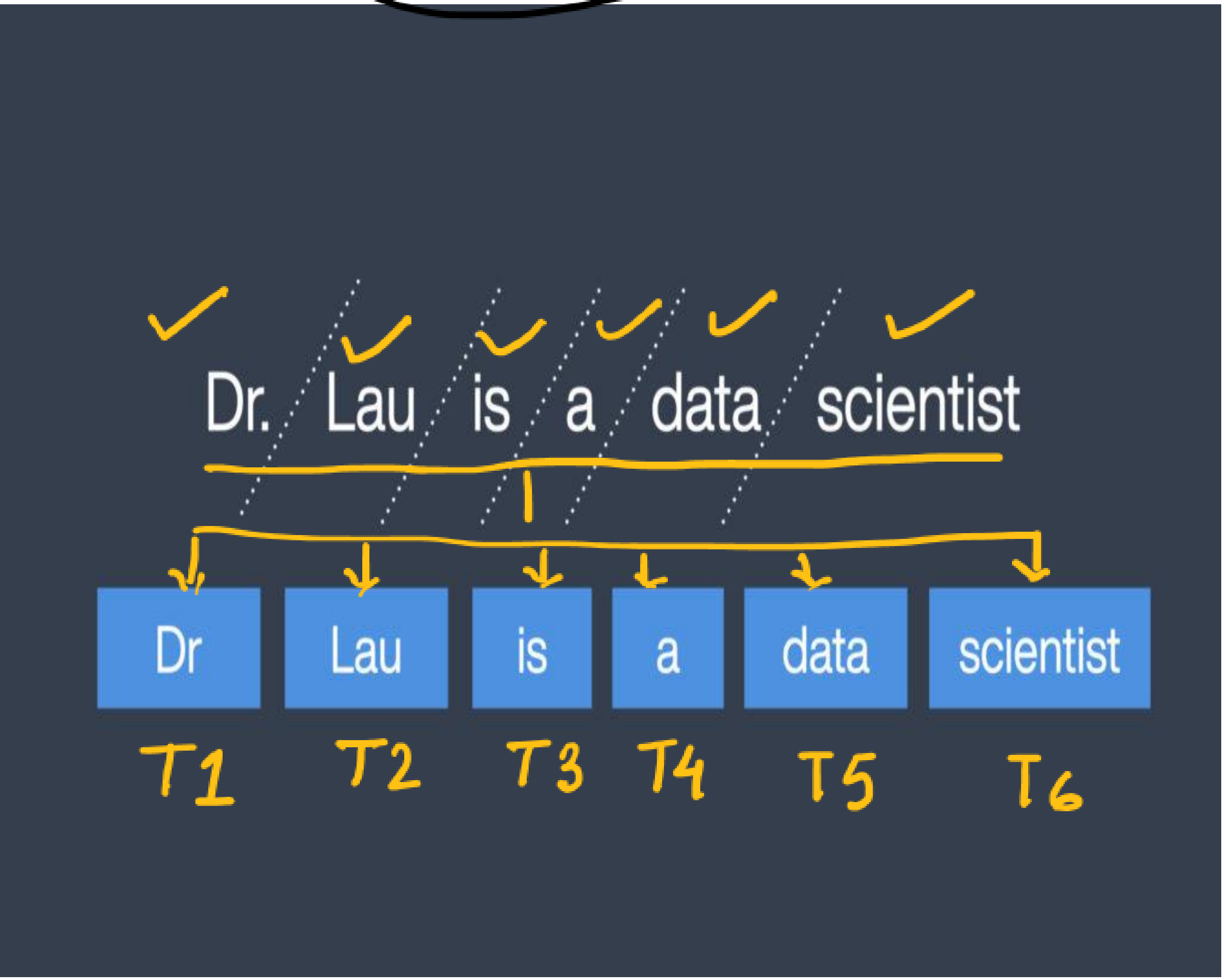
→ lemmatization

→ Stemming

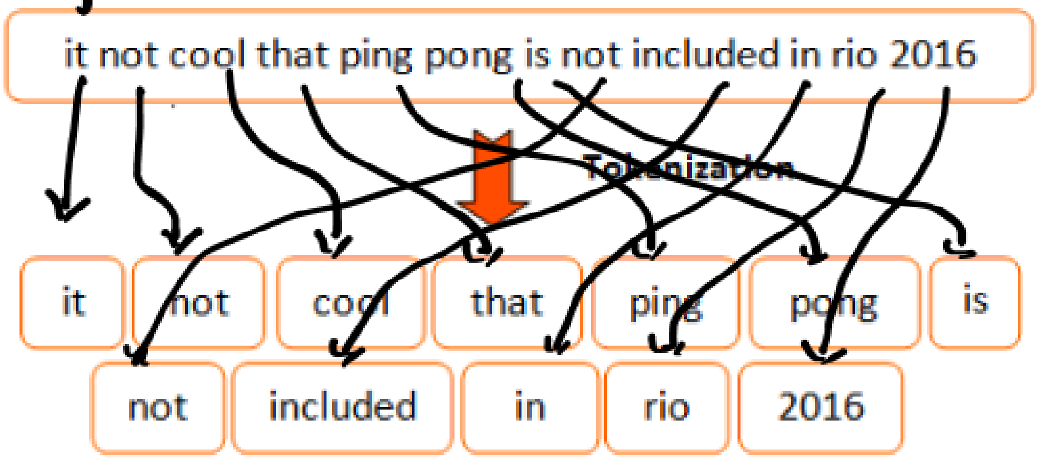
→ Stop words

Tokenisation: Splitting the sentence into words

Sentence → words (tokens)



Example :



Stopwords: → frequently occuring words but with no information, they are present in the corpus for gramatical structuring of the sentence.

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I am teaching in upgrad

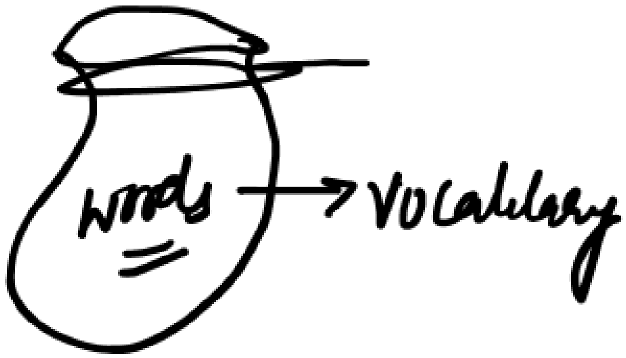
* Better to remove stop words (lexical processing)

Sample text with Stop Words	Without Stop Words
GeeksforGeeks – A Computer Science Portal for Geeks	GeeksforGeeks , Computer Science, Portal ,Geeks
Can listening be exhausting?	Listening, Exhausting
I like reading, so I read	Like, Reading, read

Text Corpus $\longrightarrow$ Words (tokens) $\longrightarrow$ Unique words (Vocabulary) $\longrightarrow$ 5 $\longrightarrow$ 2

Vocabulary $\rightarrow$ BoW

No. of unique words = No. of features



(Data become lighter)

Stemming & Lemmatization:

} Purpose of both is same

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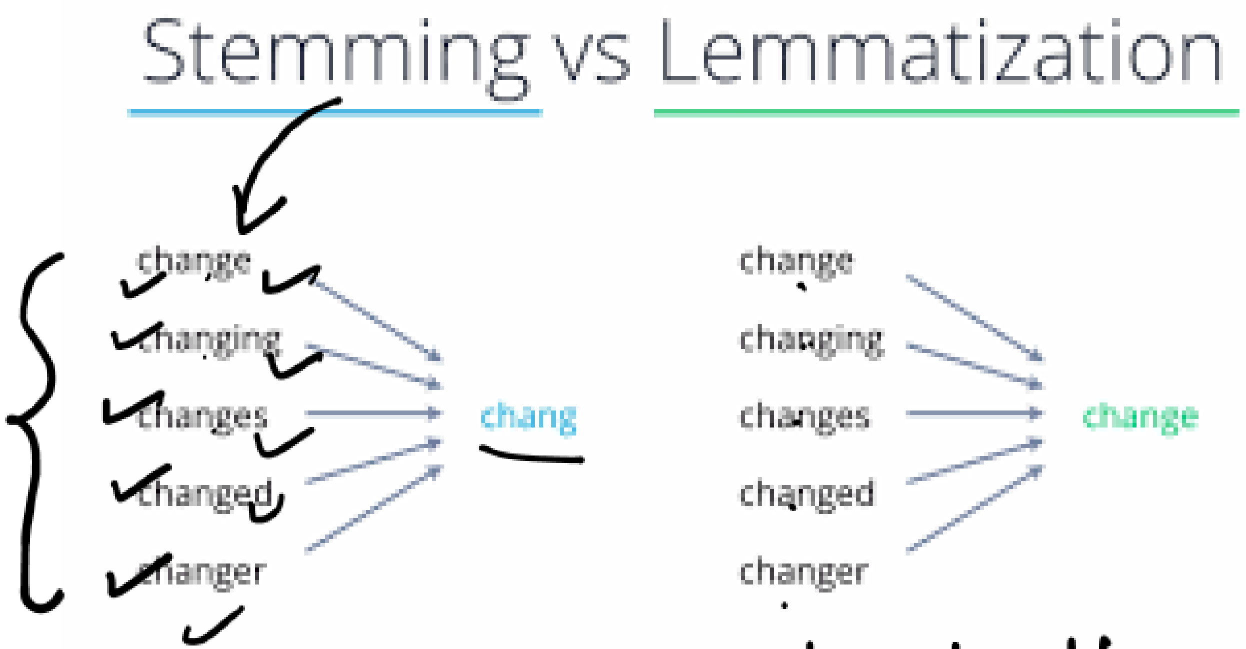
* To bring down the word at its root level. (Overall Vocablary size can be reduced)



Word	Stemming	Lemmatization
information	inform	information
informative	inform	informative
computers	comput	computer
feet	feet	foot

* Stemming is a rule based approach hence less accurate.
* Root word produced by Stemming may not be actual word.
* fast approach

→ **Lemmatization :-**

* It is database or ML approach. hence it is more accurate
* It always results actual word.
* Computationally heavier process (time consuming)

Bag of Words Representation:

(Naive Method)

(NLP) → Text data → Numerical data → ML task

(Vectors)

Advanced NLP

* BoW Method
* TF-IDF Method

7 unique words → 7 features → data

Unique words	about	bird	heard	is	the	word	you
D1 About the bird, the bird, bird bird bird	1	5	0	0	2	0	0
D2 You heard about the bird	1	1	1	0	1	0	1
D3 The bird is the word	0	1	0	1	2	1	0

$$
\begin{tikzcd}
D1 \arrow[r] & {[1,5,0,0,2,0,0]} \arrow[d] & \\
\text{Text data} \arrow[u] \arrow[r, bend left, "\text{BoW}"'] & \text{Vector} \arrow[r] & \text{ML task}
\end{tikzcd}
$$

TF-IDF:

importance of words in Document

= Term frequency x inverse document frequency

importance of words in corpus

$f_{t,d}$

	blue	bright	can	see	shining	sky	sun	today
1	1	0	0	0	0	1	0	0
2	0	1	0	0	0	0	1	1
3	0	1	0	0	0	1	1	0
4	0	1	1	1	1	0	2	0

$$\mathtt{tf}(t,d) = \frac{f_{t,d}}{\sum_{t'} f_{t',d}}$$

	blue	bright	can	see	shining	sky	sun	today
1	1/2	0	0	0	0	1/2	0	0
2	0	1/3	0	0	0	0	1/3	1/3
3	0	1/3	0	0	0	1/3	1/3	0
4	0	1/6	1/6	1/6	1/6	0	1/3	0

$f_{t,d}$

	blue	bright	can	see	shining	sky	sun	today
1	1	0	0	0	0	1	0	0
2	0	1	0	0	0	0	1	1
3	0	1	0	0	0	1	1	0
4	0	1	1	1	1	0	2	0
n_t	**1**	**3**	**1**	**1**	**1**	**2**	**3**	**1**

$N = 4$

$$\mathtt{idf}(t,D) = \log_{10} \frac{N}{n_t}$$

	blue	bright	can	see	shining	sky	sun	today
	0.602	0.125	0.602	0.602	0.602	0.301	0.125	0.602

$\log_{10} \frac{4}{1} = 0.602$ $\log_{10} \frac{4}{3} = 0.125$

→ **Term frequency** :-

$$f_t = \frac{\text{No. of times that word occured in document}}{\text{Total no. of words in document}}$$

→ **Inverse document frequency** :-

$$f_d = \log \frac{\text{No. of document in which that word is coming } (d)}{\text{Total No of document } (D)}$$

$$\boxed{f_{id} = \log \frac{D}{d}}$$

Term frequency	about	bird	heard	is	the	word	you
About the bird, the bird, bird bird bird	1/8	5/8	0/8	0/8	2/8	0/8	0/8
You heard about the bird	1/5	1/5	1/5	0/5	1/5	0/5	1/5
The bird is the word	0/5	1/5	0/5	1/5	2/5	1/5	0/5

IDF	about	bird	heard	is	the	word	you
About the bird, the bird, bird bird bird	1 $\log \frac{3}{2}$	5 $\log \frac{3}{3}$	0 $\log \frac{3}{1}$	0 $\log \frac{3}{1}$	2	0	0
You heard about the bird	1	1	1	0	1	0	1
The bird is the word	0	1	0	1	2	1	0

TF-IDF:

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$\mathtt{tf}(t, d)$

	blue	bright	can	see	shining	sky	sun	today
1	1/2	0	0	0	0	1/2	0	0
2	0	1/3	0	0	0	0	1/3	1/3
3	0	1/3	0	0	0	1/3	1/3	0
4	0	1/6	1/6	1/6	1/6	0	1/3	0

x

$\mathtt{idf}(t, D)$

	blue	bright	can	see	shining	sky	sun	today
	0.602	0.125	0.602	0.602	0.602	0.301	0.125	0.602

$$\mathtt{tfidf}(t, d, D) = \mathtt{tf}(t, d) \times \mathtt{idf}(t, D)$$

	blue	bright	can	see	shining	sky	sun	today
1	**0.301**	0	0	0	0	0.151	0	0
2	0	0.0417	0	0	0	0	0.0417	**0.201**
3	0	0.0417	0	0	0	**0.100**	0.0417	0
4	0	0.0209	**0.100**	**0.100**	**0.100**	0	0.0417	0

- TF-IDF: Multiply TF and IDF scores, use to rank importance of words within documents
  - Most important word for each document is highlighted

PMI: (Advanced tokenization)

Point-wise mutual information (PMI)

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Single token

New delhi

T1 T2

Token

OCCURRENCE CONTEXT

Corpus
S1 : Constraction of a new metro station state in New Delhi
S2 : New Delhi is the capital of India
S3 : The climate of Delhi is monsoon-influenced humid climate

$$p(\text{New Delhi}) = \frac{2}{3}$$

$$p(\text{New}) = \frac{2}{3}$$

$$p(\text{Delhi}) = \frac{3}{3}$$

$$pmi(\text{New Delhi}) = \frac{p(\text{New Delhi})}{p(\text{New})\,p(\text{Delhi})} = \frac{\text{Prob. of combined word}}{\text{Product of Prob. of individual words}}$$

$$PMI = \frac{J.P}{M.P} \qquad \frac{2/3}{2/3 \times 1} = 1$$

PMI > 1 → Tokens should be considered as combined

PMI ≥ 1
(few textbooks)

$J \cdot P > M \cdot P$

PMI < 1 → Tokens should be considered separately only

$J \cdot P < M \cdot P$

PMI = 1 → Dicy situation (Call can be taken based on Use-case)

$J \cdot P = M \cdot P$

Syntactic Processing

Branch of NLP where we deal with syntax of the language.

→ intermediate }

→ Grammatical Structure of the sentence

* Order / Sequence matters.
* Grammatical structure matters.
* Do not remove the stop words
* Context matters

PoS Tagging: Part-of-speech tagging

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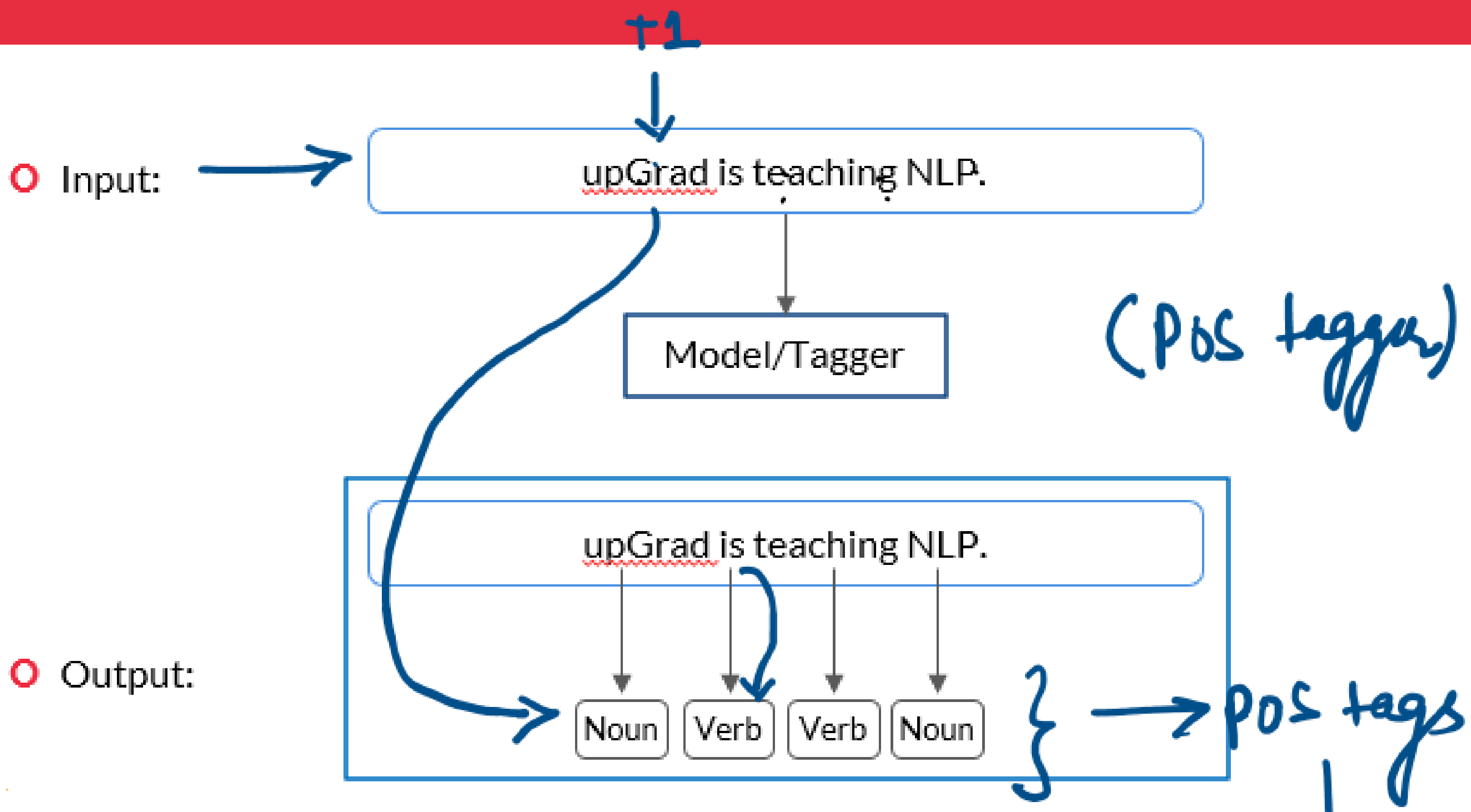

Used as a feature

Parsing: Helps us to understand the complex grammatical structure of the sentence.

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Sentence (complex) → Tree (Simpler) → The quick brown fox jumps over the table.

The quick brown fox → Noun Phrase

jumps over the table → Verb phrase

* Constituency Parsing :-

S1

NP — The quick brown fox

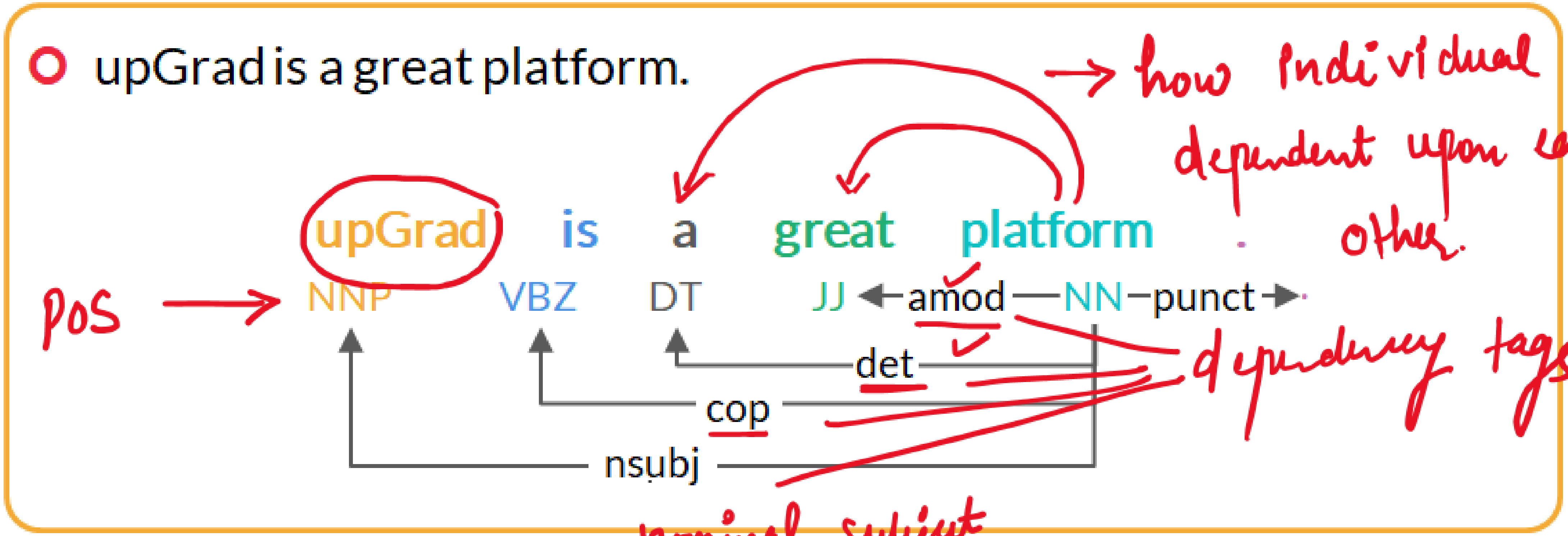
VP

- VP → jumps
- PP → over the table

Verb phrase

- jumps (VP)
- over the table (PP)

→ Dependency parsing :- (helps us to understand the context of words with the relationships (dependencies) with the surrounding words.



→ how individual tokens dependent upon each other.

POS →

dependency tags

nominal subject

NER:{ Name - entity - Recognition

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→ key elements present in the text

→ Nouns

→ Book a {flight} from {bangalore} to {Delhi} on {friday}

* Name of person

* Name of place

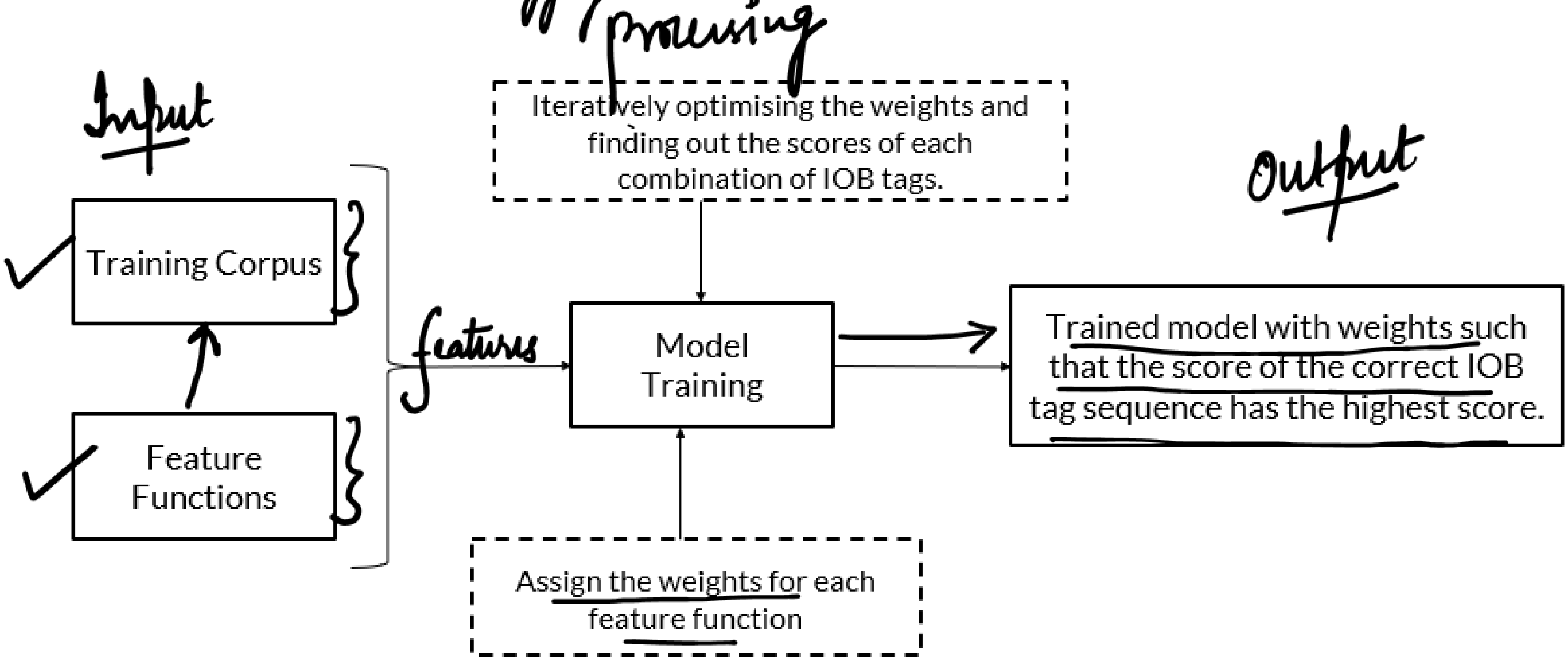
* Name of organisation

* Number

NER → majority information lies in NER

NER & CRF: Conditional Random fields

IOB Tagging → NER



Semantic Processing:(NLP Branch)

Complete context of the sentence

- Knowledge Graph (KG)
- Topic modelling

King → V_1

Queen → V_2

girl → V_3

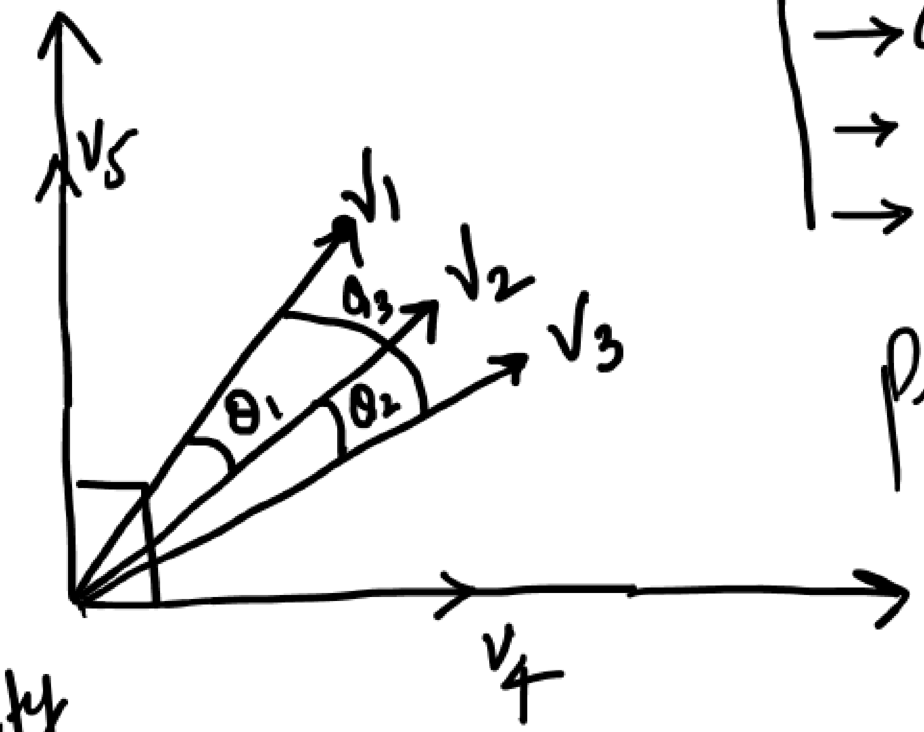
intuitive

↑

cosine similarity

$\theta_1 < \theta_3$

$(V_1 \& V_2) > (V_1 \& V_3)$



- → Glove Embed
- → Bert Embed
- → Lstm Embed

Pre-trained

$\cos 90 = 0$

Word2Vec

- → Glove Embed
- → Bert Embed
- → Lstm Embed

My ______

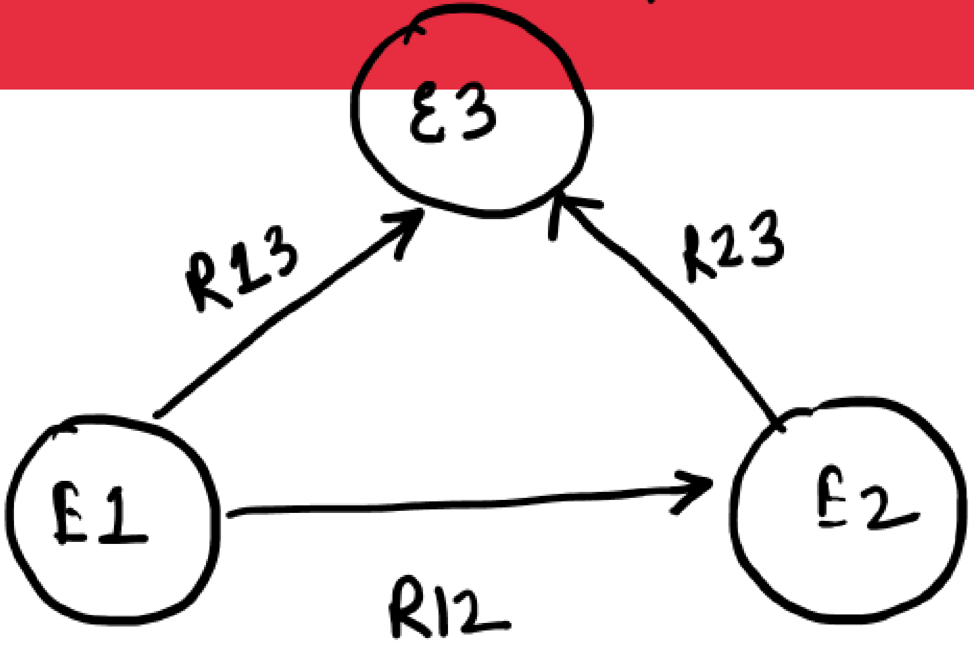
My name

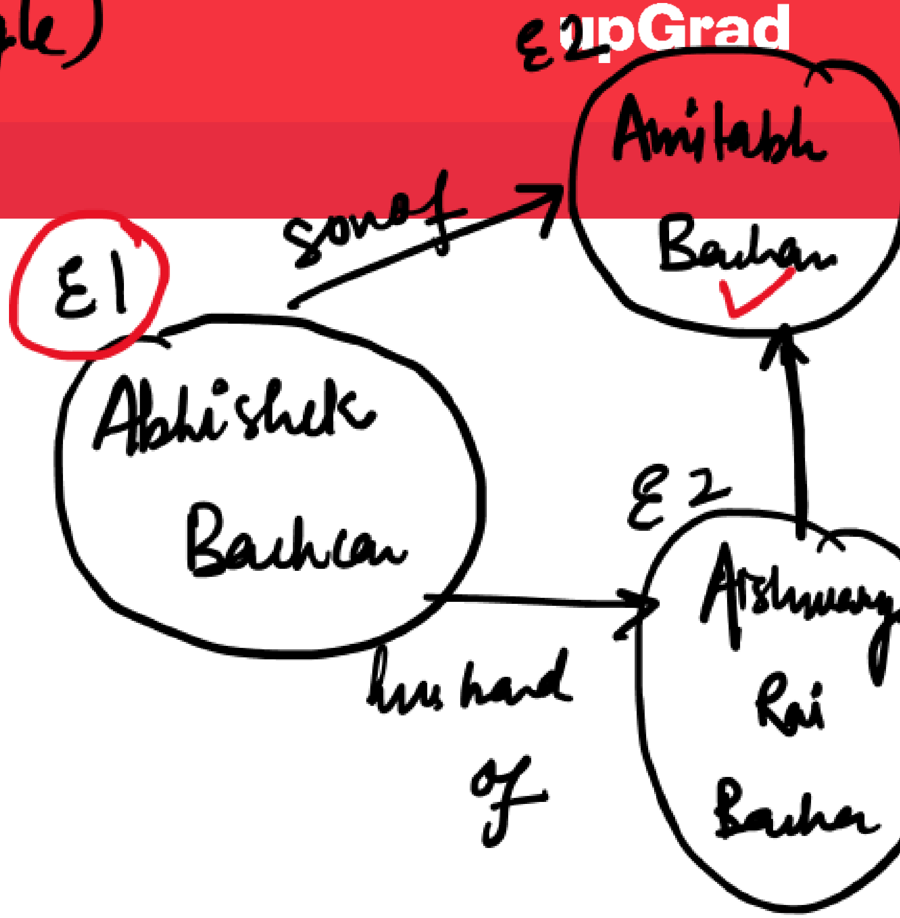
List after my

W1	10	$\frac{10}{20}$
W2	2	$\frac{2}{20}$
W3	2	$\frac{2}{20}$
W4	3	$\frac{3}{20}$
W5	3	$\frac{3}{20}$

$\underline{\underline{20}}$

Knowledge Graphs: (pioneered by google)





Wordnet → database which is used for knowledge graph generation

Topic Modelling: (Unsupervised ML Technique)

* Target variable is missing

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TOPIC MODELLING

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Topic 1	Model, data, validation, neural networks, supervised, machine learning, etc.
Topic 2	Cancer, hospitals, medicine, X-ray, disease, etc.
Topic 3	Chef, restaurants, chocolate, spicy, food, etc.

→ T1
→ T2
T3

Document 1: Machine learning can aid and improve early detection of breast cancer → Topic

Document 1 talks about Topic 1 and Topic 2 → → Topic

Doubts:

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